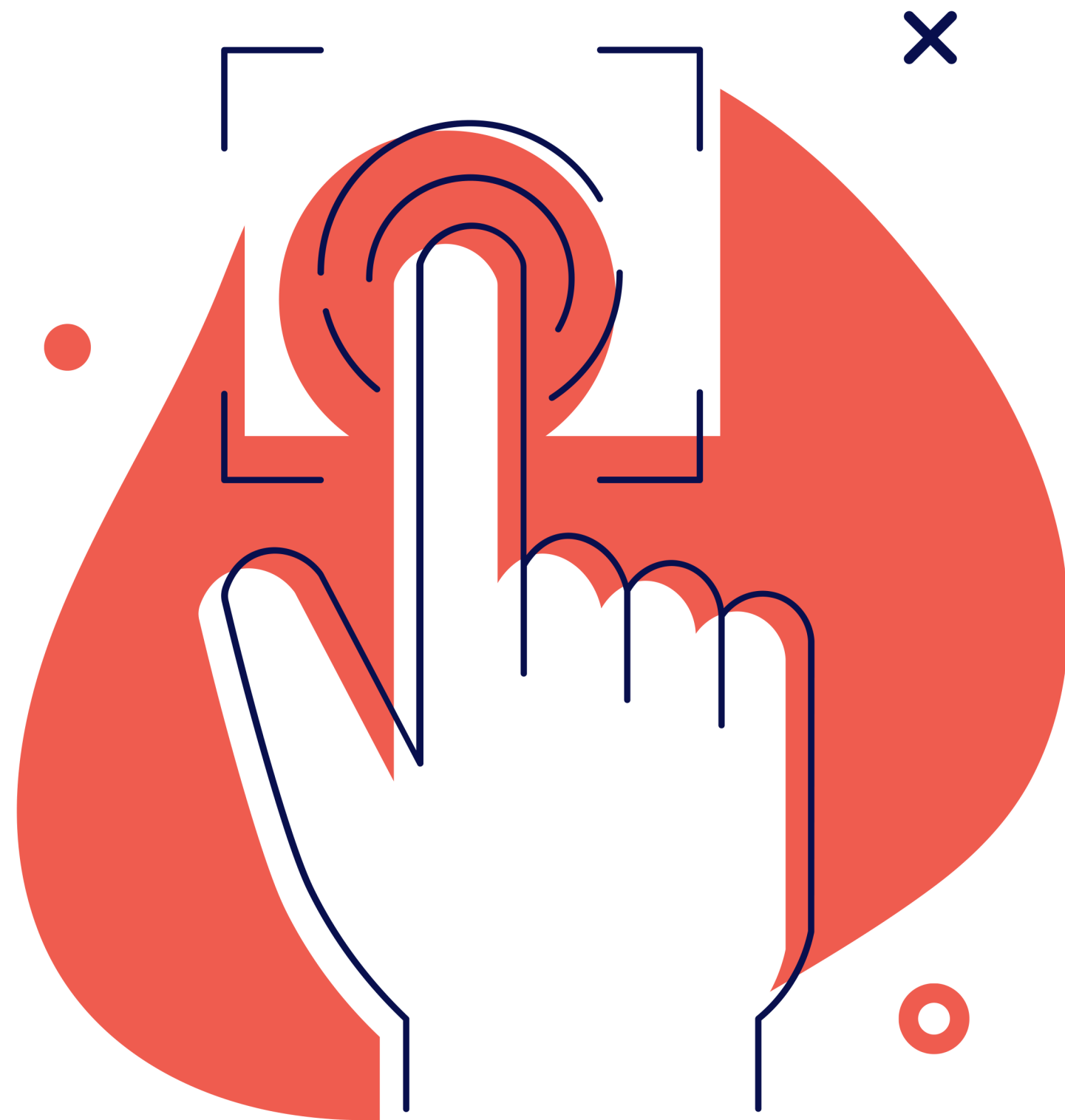




# How to create an AWS EKS Cluster

Backend Master Class #30



# What is Kubernetes?



# What is Kubernetes?

- An open-source container orchestration engine



# What is Kubernetes?

- An open-source container orchestration engine
- For automating deployment, scaling, and management of containerized applications





# Kubernetes' components



# Kubernetes' components



Worker Node



# Kubernetes' components



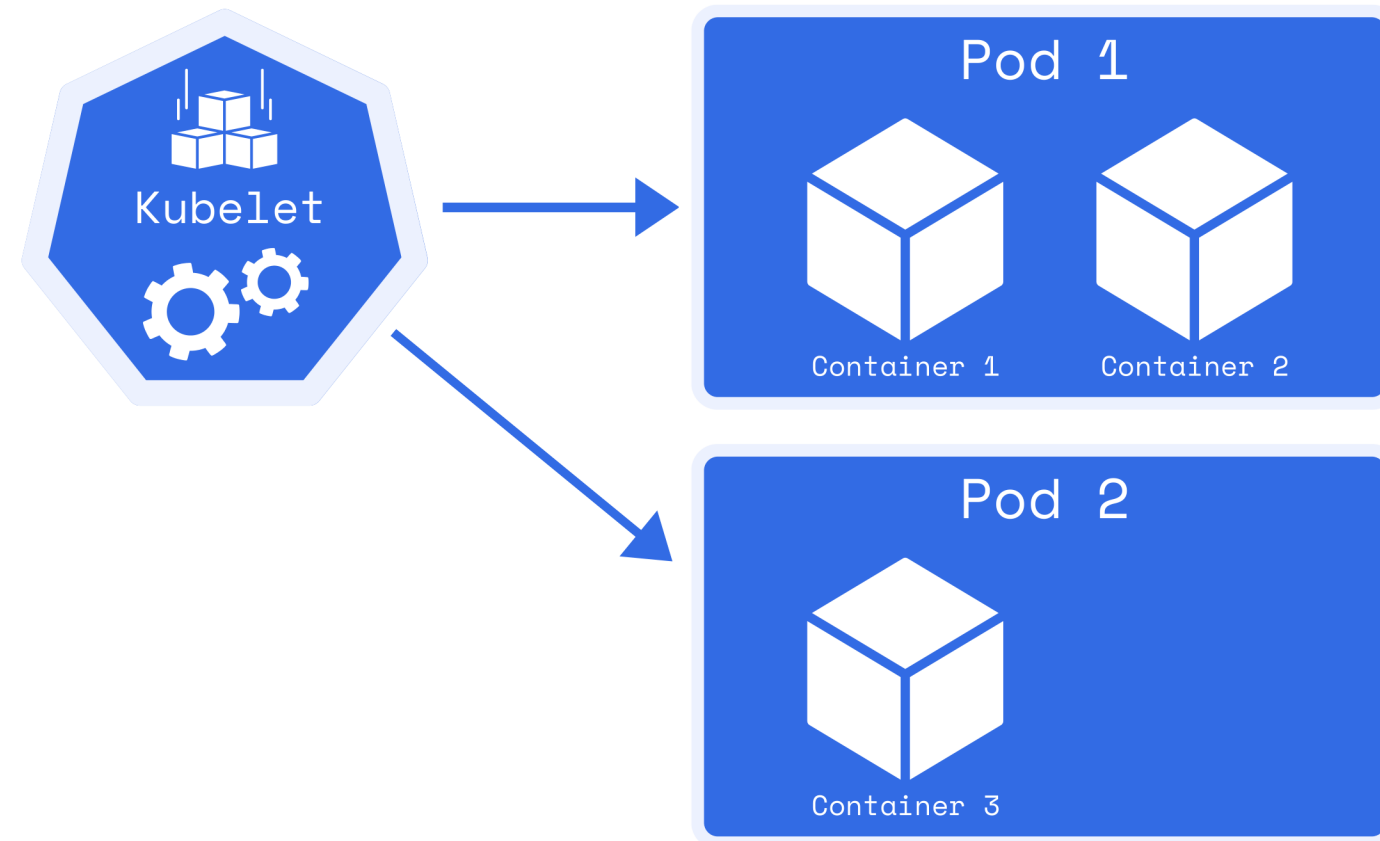
Worker Node



# Kubernetes' components



## Worker Node



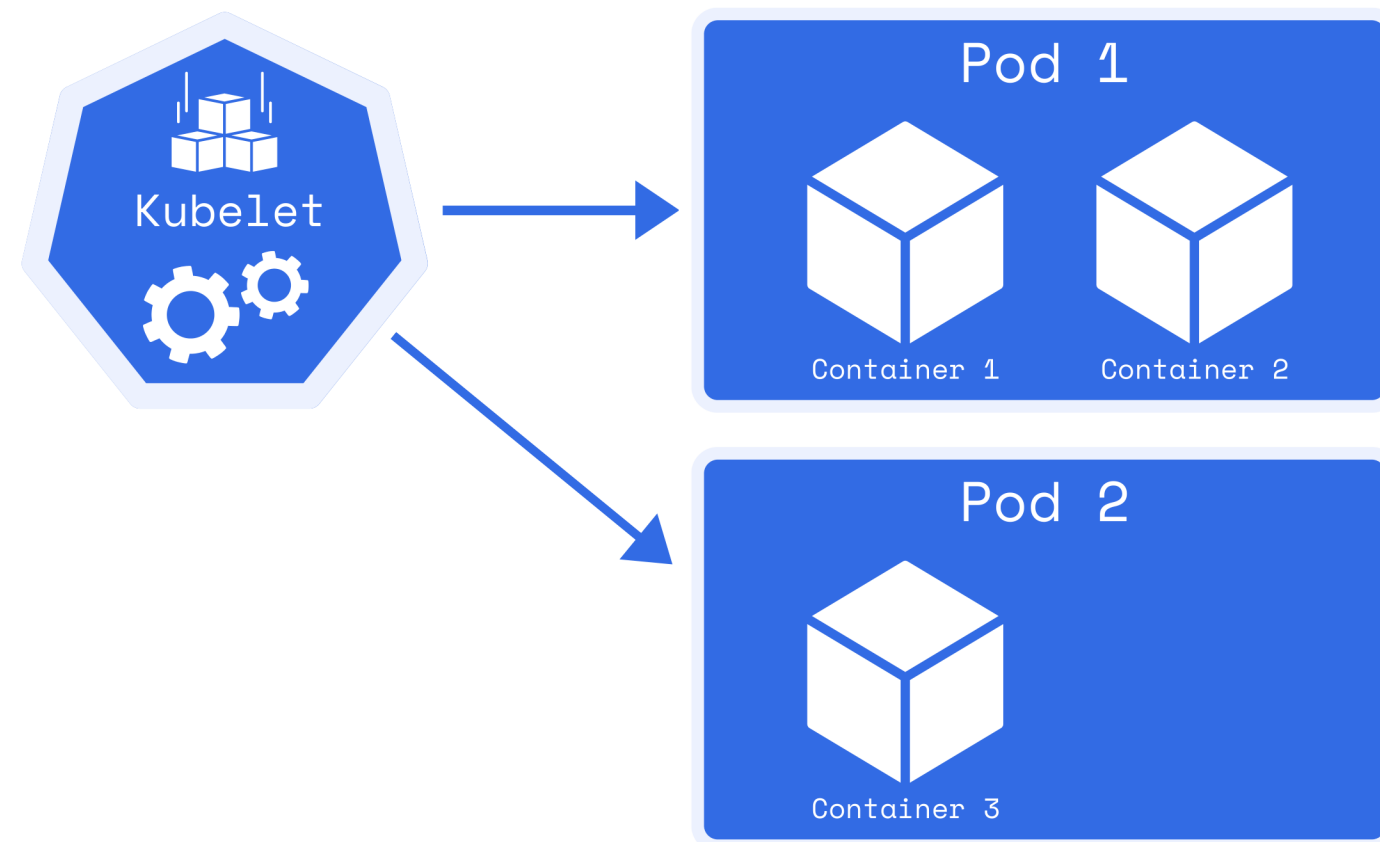
- Kubelet agent: make sure containers run inside pods



# Kubernetes' components



## Worker Node



- Kubelet agent: make sure containers run inside pods
- Container runtimes: Docker, containerd, CRI-O

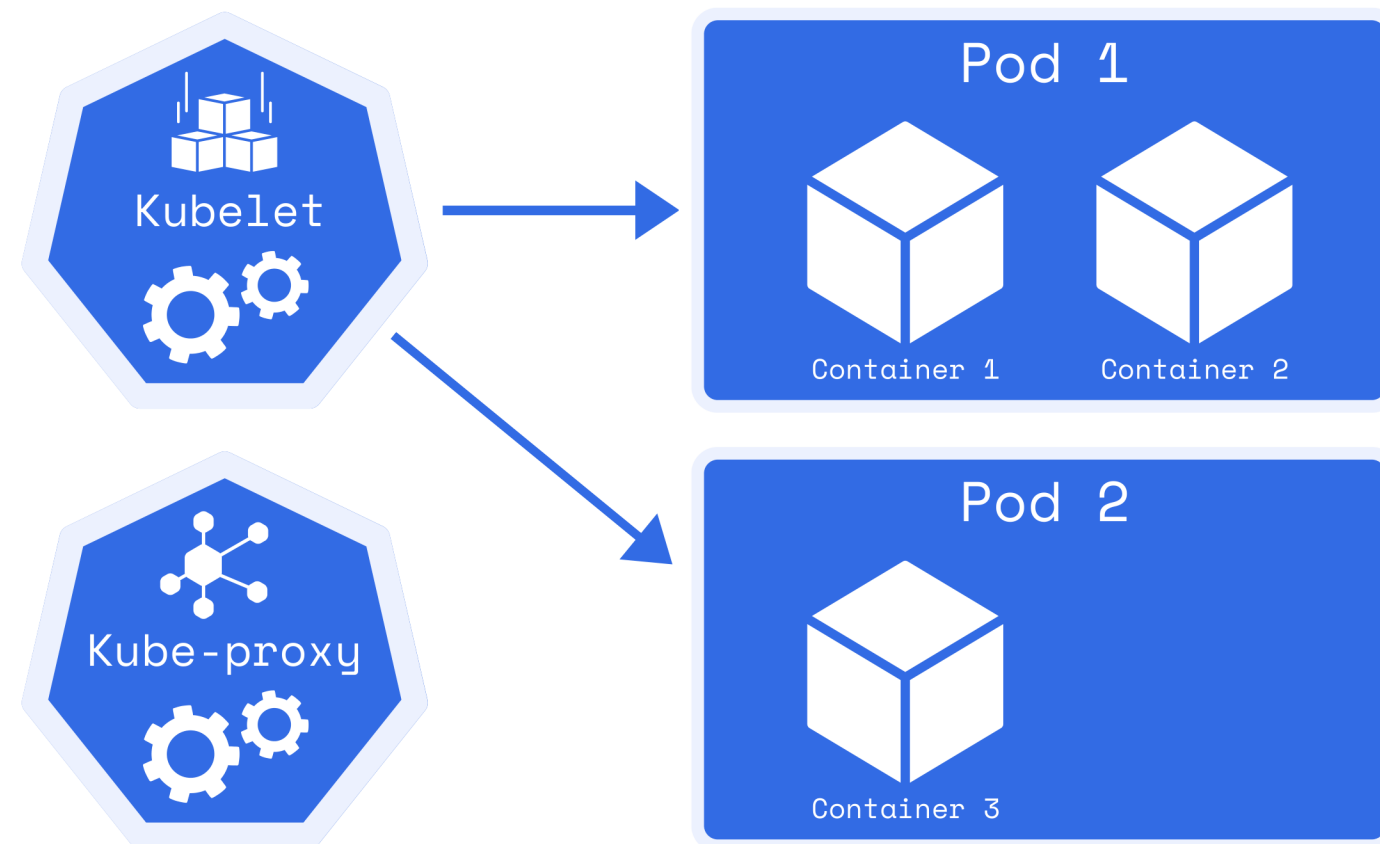




# Kubernetes' components



## Worker Node



- Kubelet agent: make sure containers run inside pods
- Container runtimes: Docker, containerd, CRI-O
- Kube-proxy: maintain network rules, allow communication with pods



# Kubernetes' components



Master Node

Control plane



# Kubernetes' components

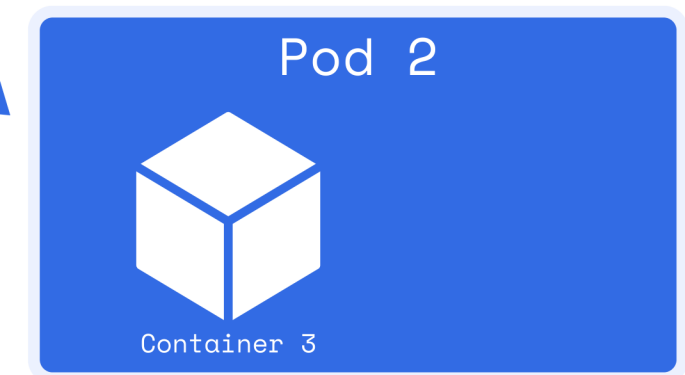
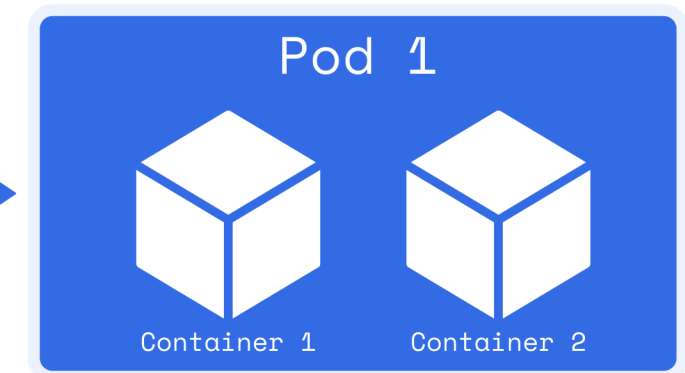
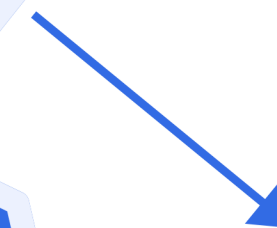


Master Node

Control plane



Worker Node





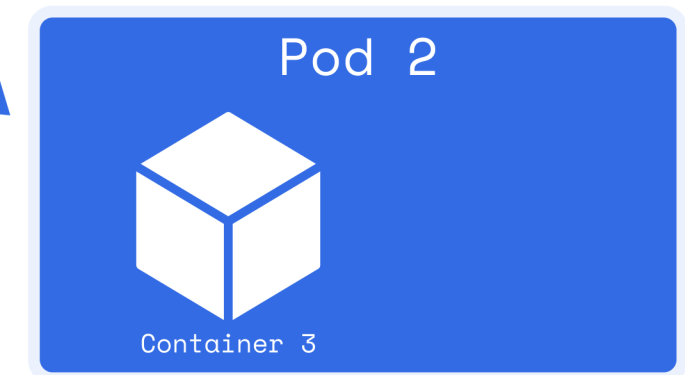
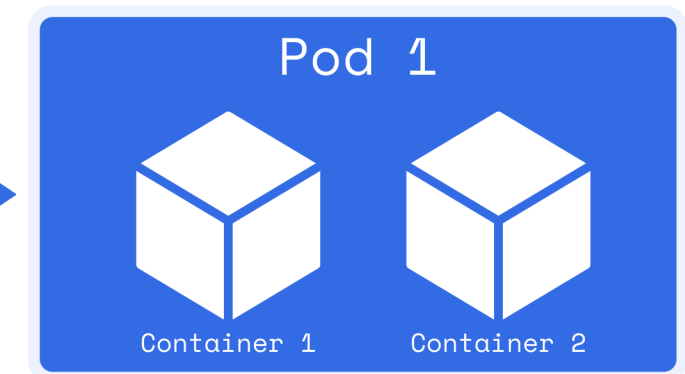
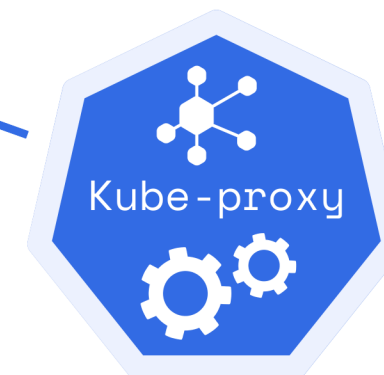
# Kubernetes' components



Master Node



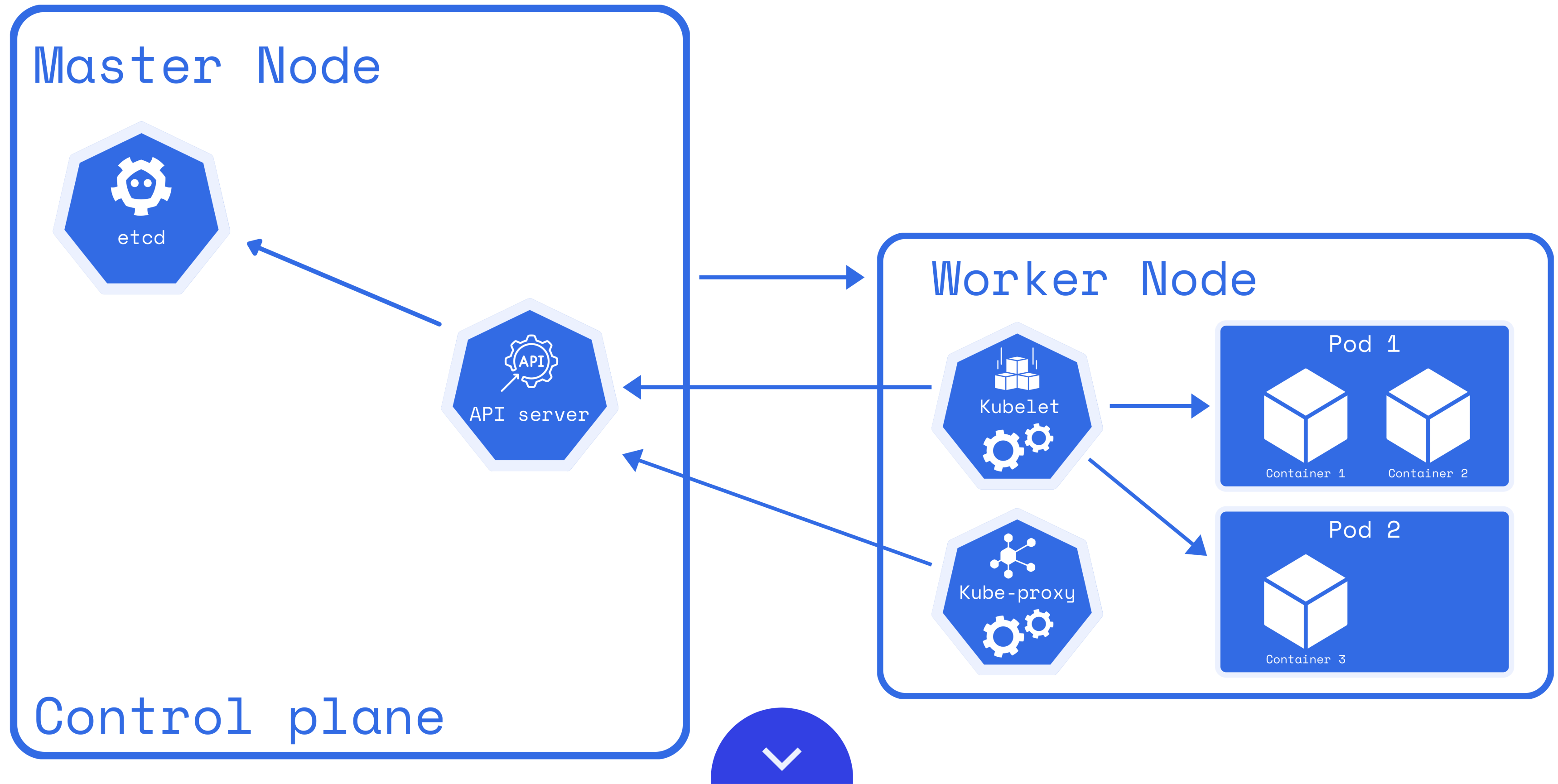
Worker Node



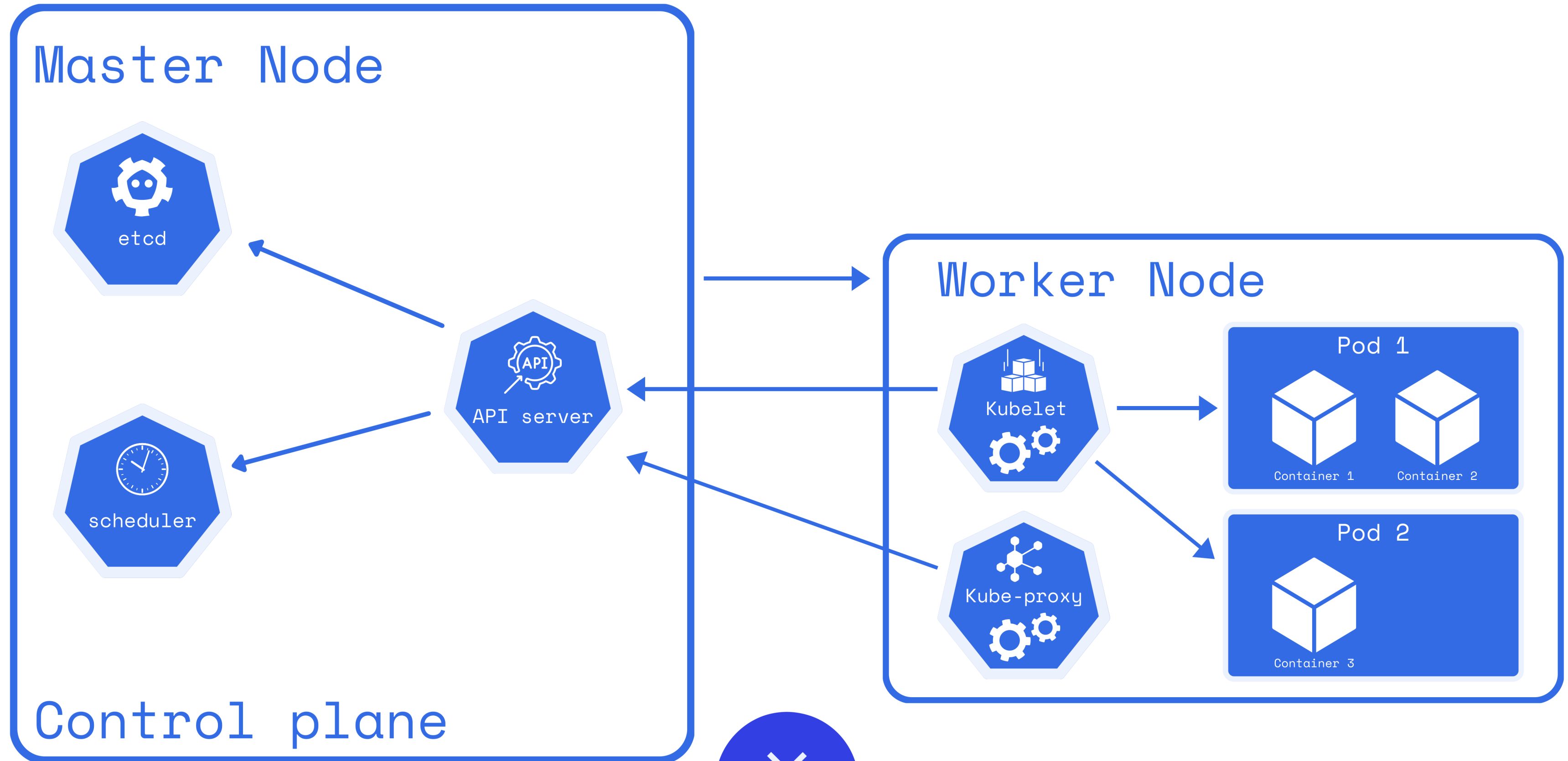
Control plane



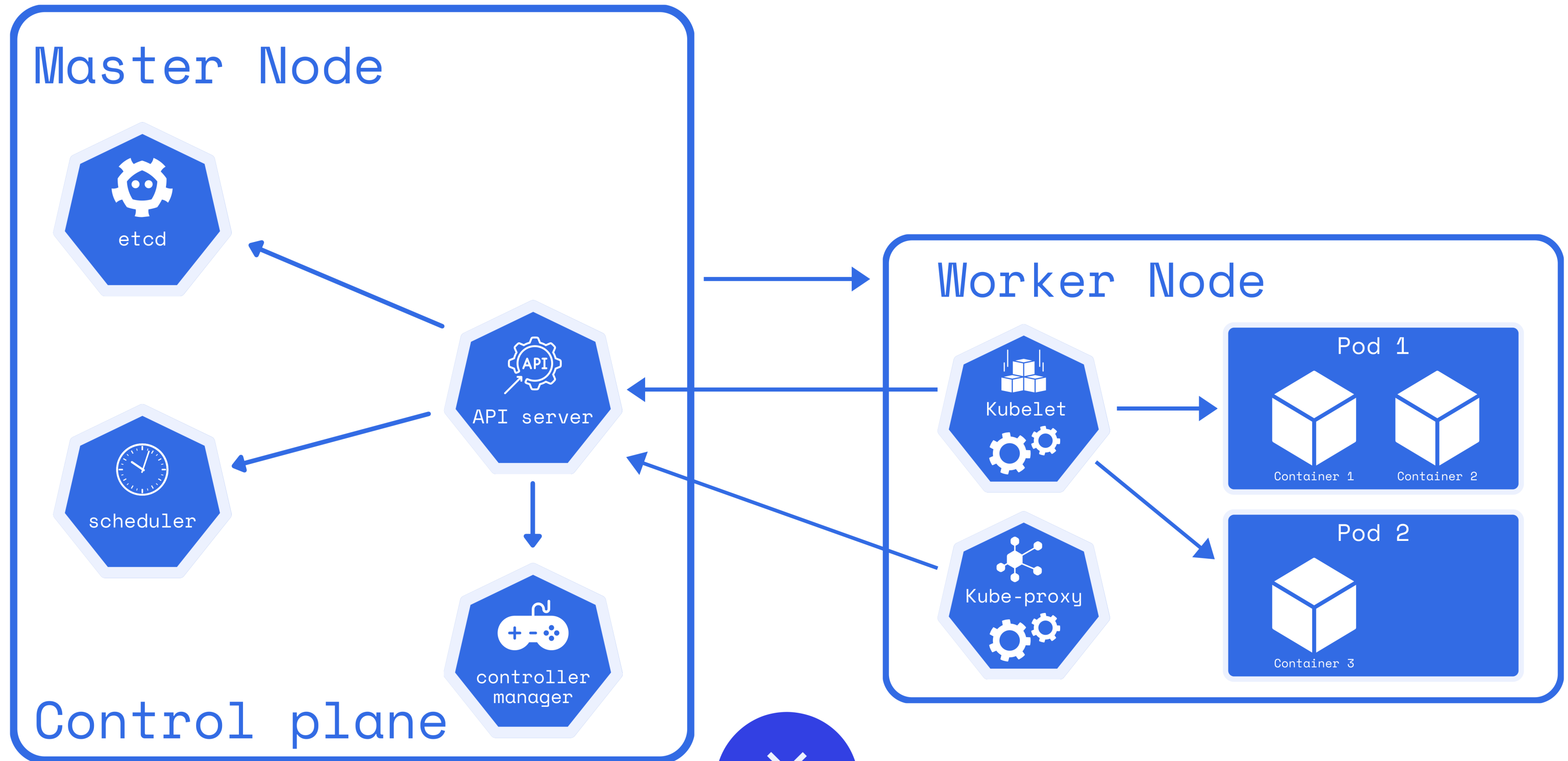
# Kubernetes' components



# Kubernetes' components



# Kubernetes' components



# Kubernetes' components



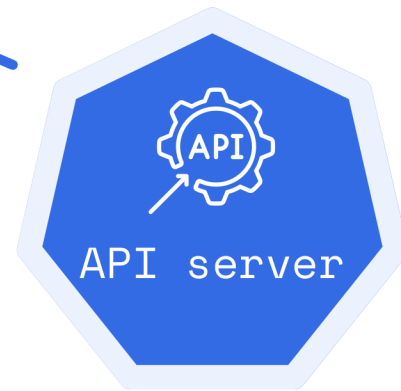
Master Node



etcd



scheduler



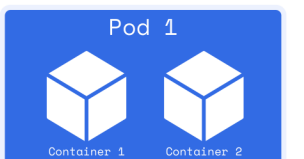
API server



controller manager

Control plane

Worker Node

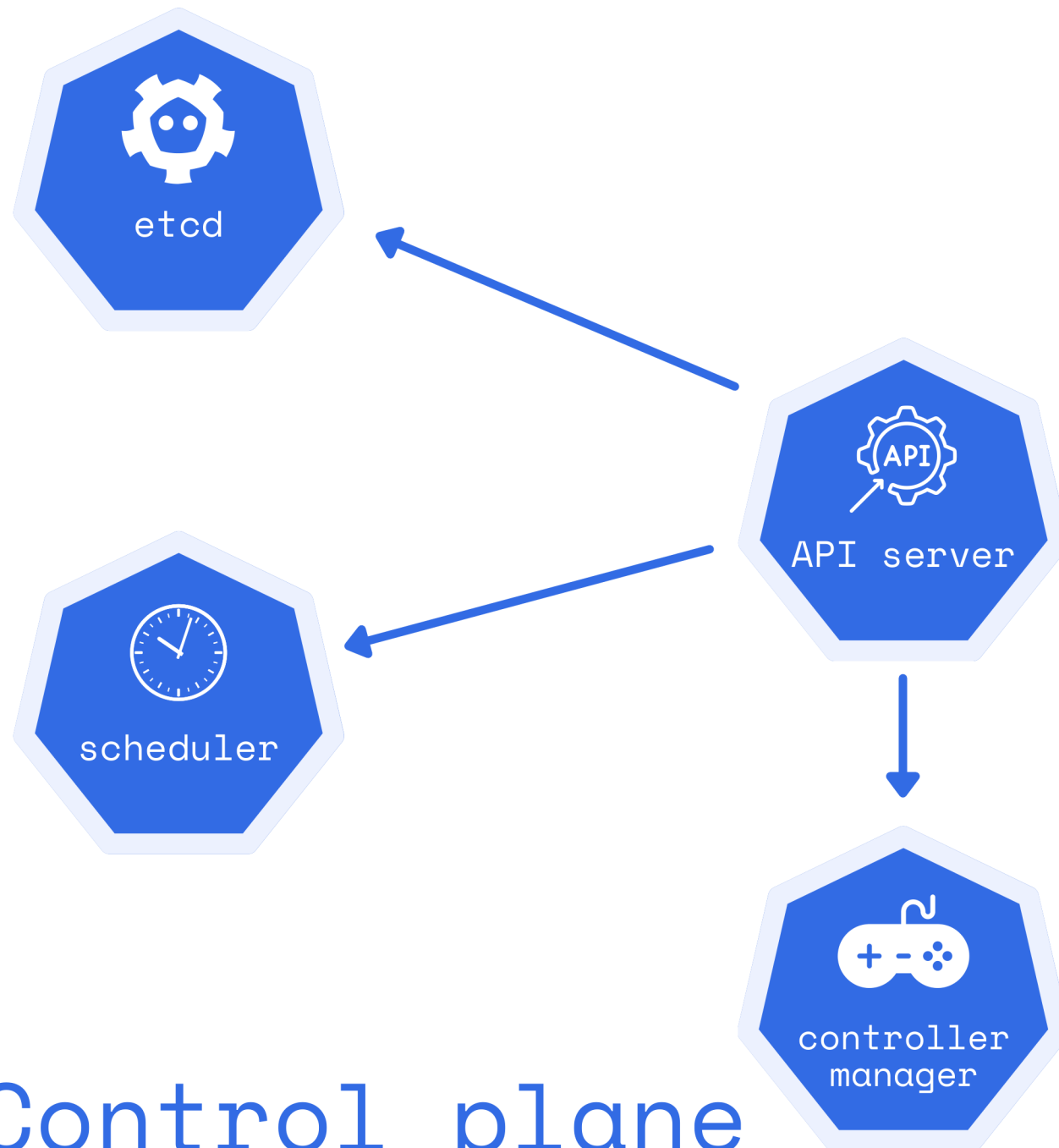




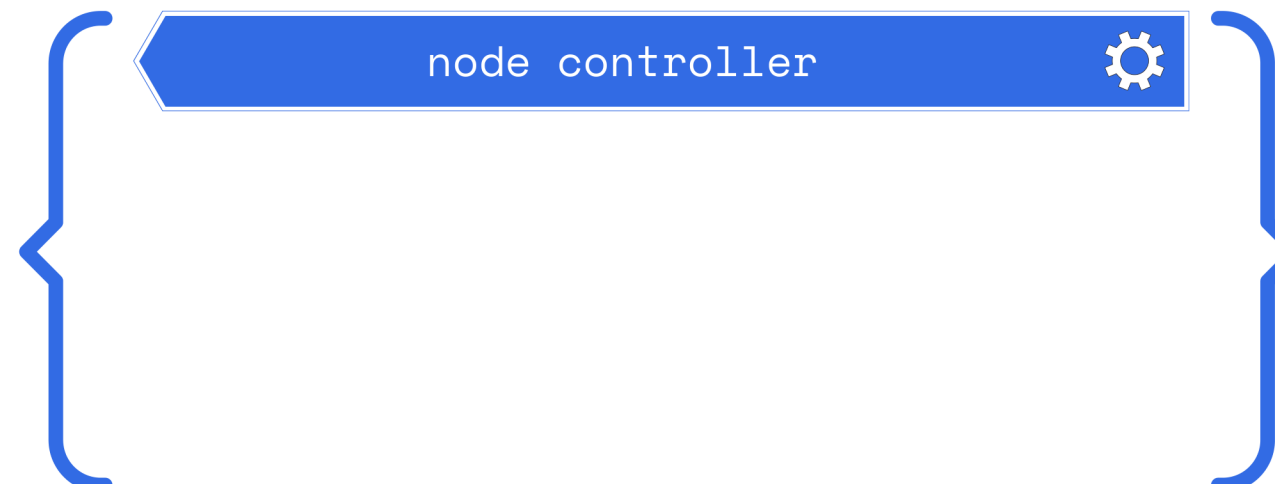
# Kubernetes' components



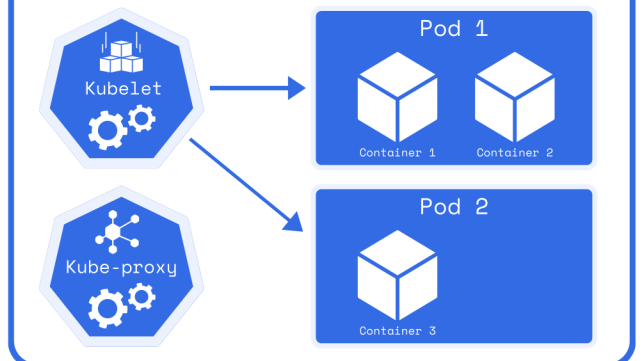
## Master Node



## Control plane



## Worker Node



# Kubernetes' components



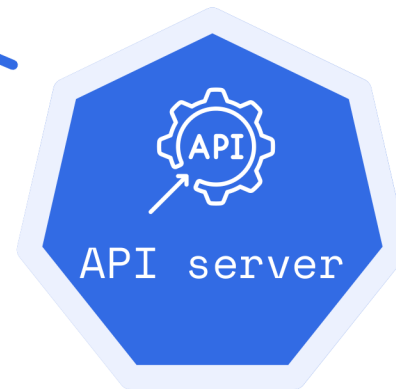
## Master Node



etcd



scheduler



API server



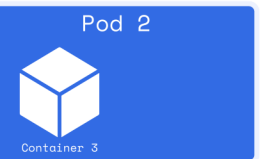
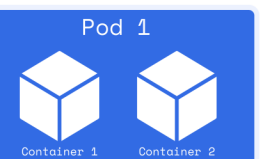
controller manager

node controller

job controller

Control plane

## Worker Node



Pod 1

Container 1 Container 2

Pod 2

Container 3

# Kubernetes' components



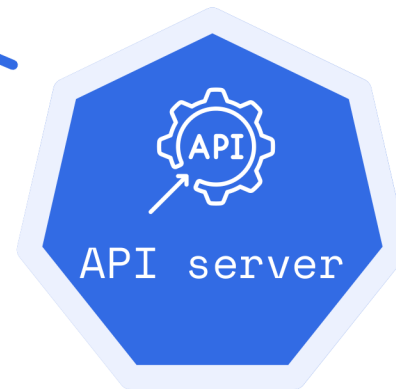
## Master Node



etcd



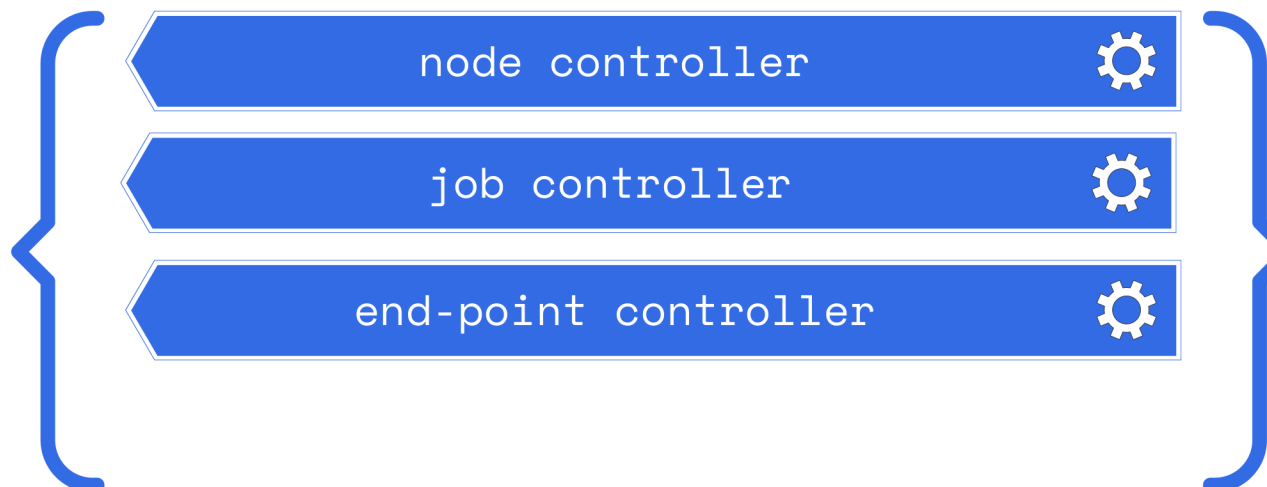
scheduler



API server

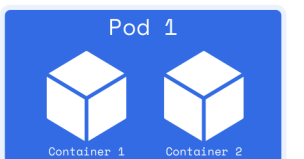


controller manager



## Control plane

## Worker Node



Pod 1

Container 1 Container 2

Pod 2

Container 3



# Kubernetes' components



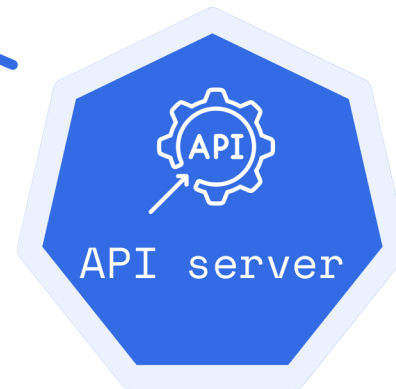
Master Node



etcd



scheduler



API server



controller manager

node controller

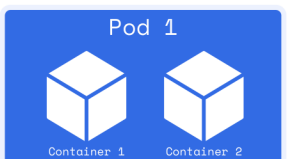
job controller

end-point controller

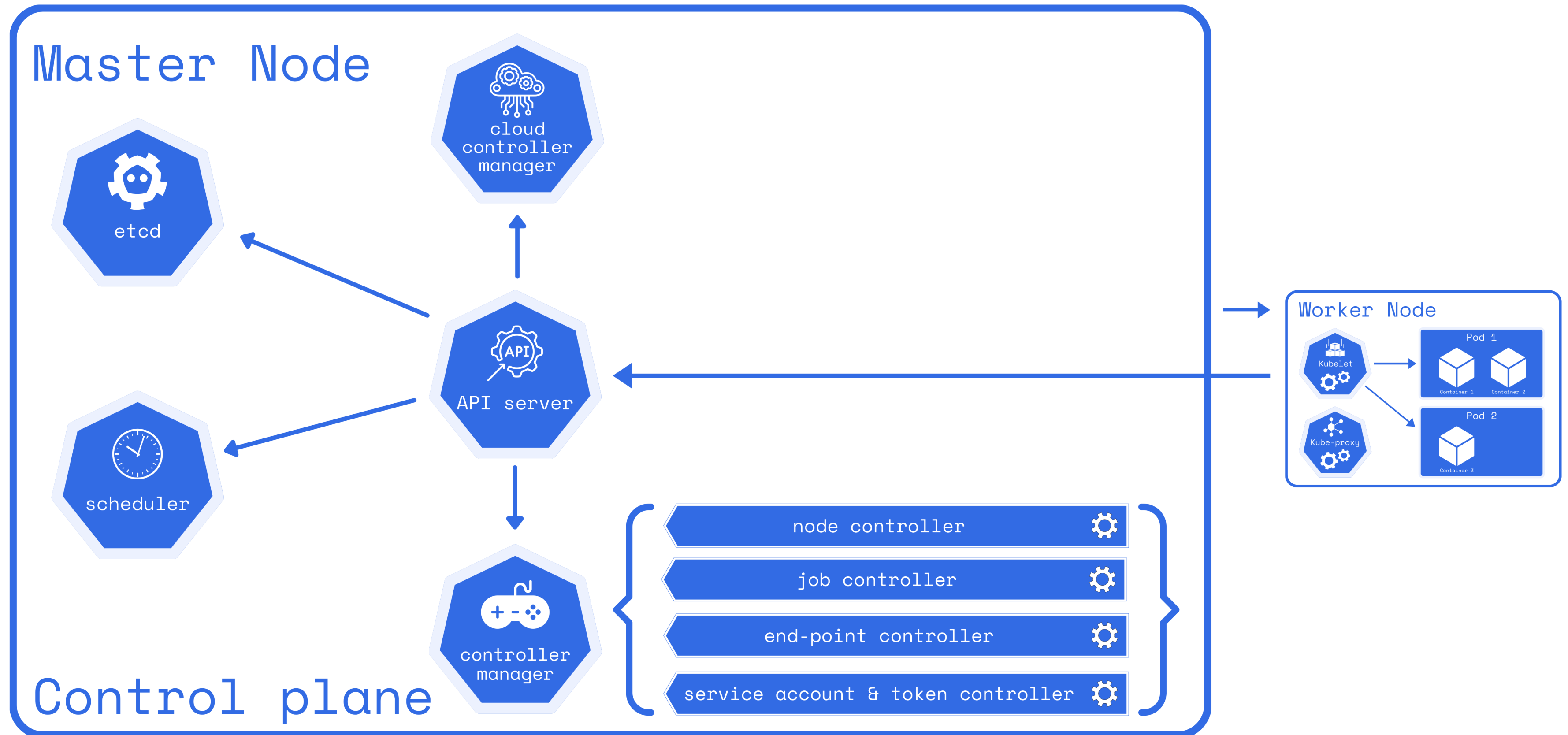
service account & token controller

Control plane

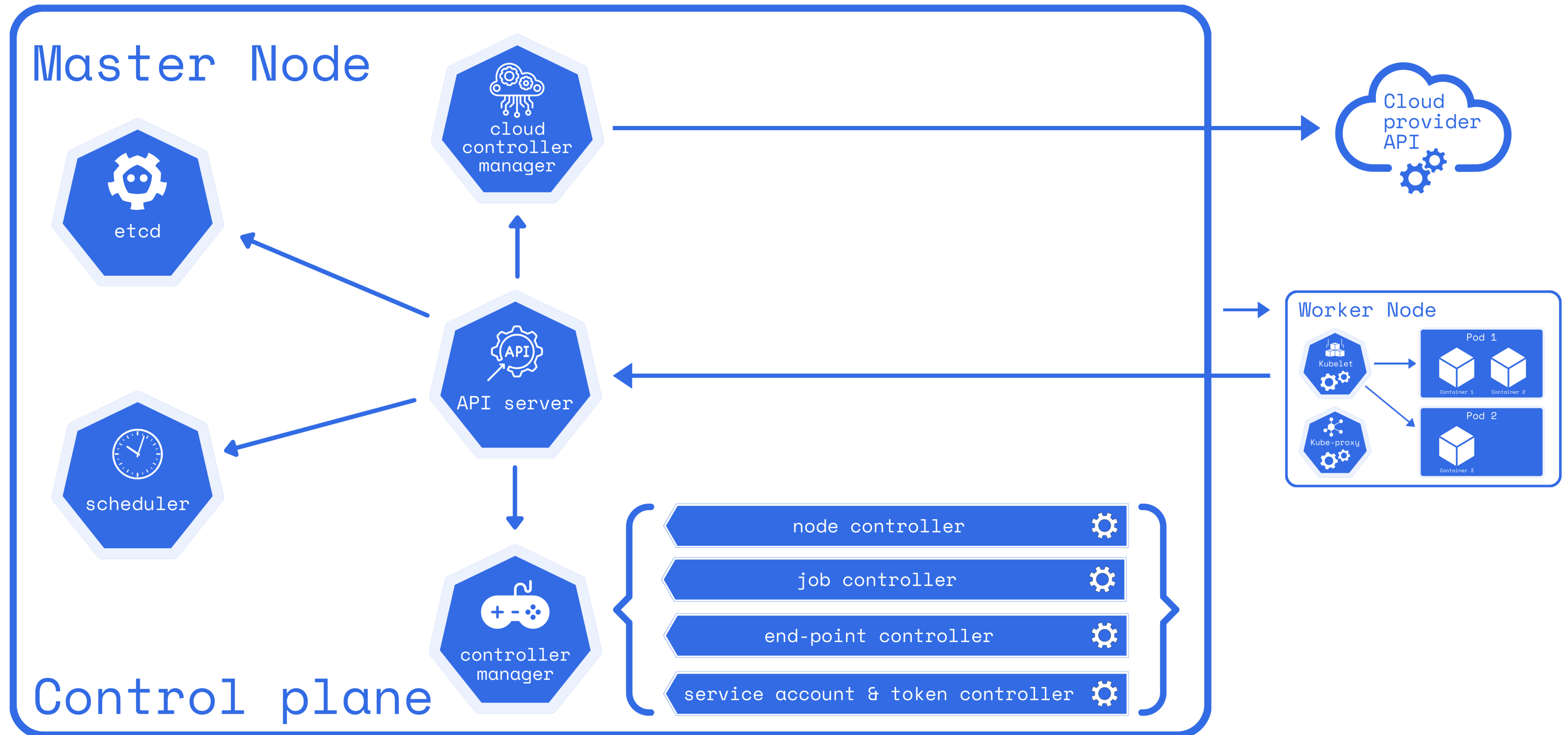
Worker Node



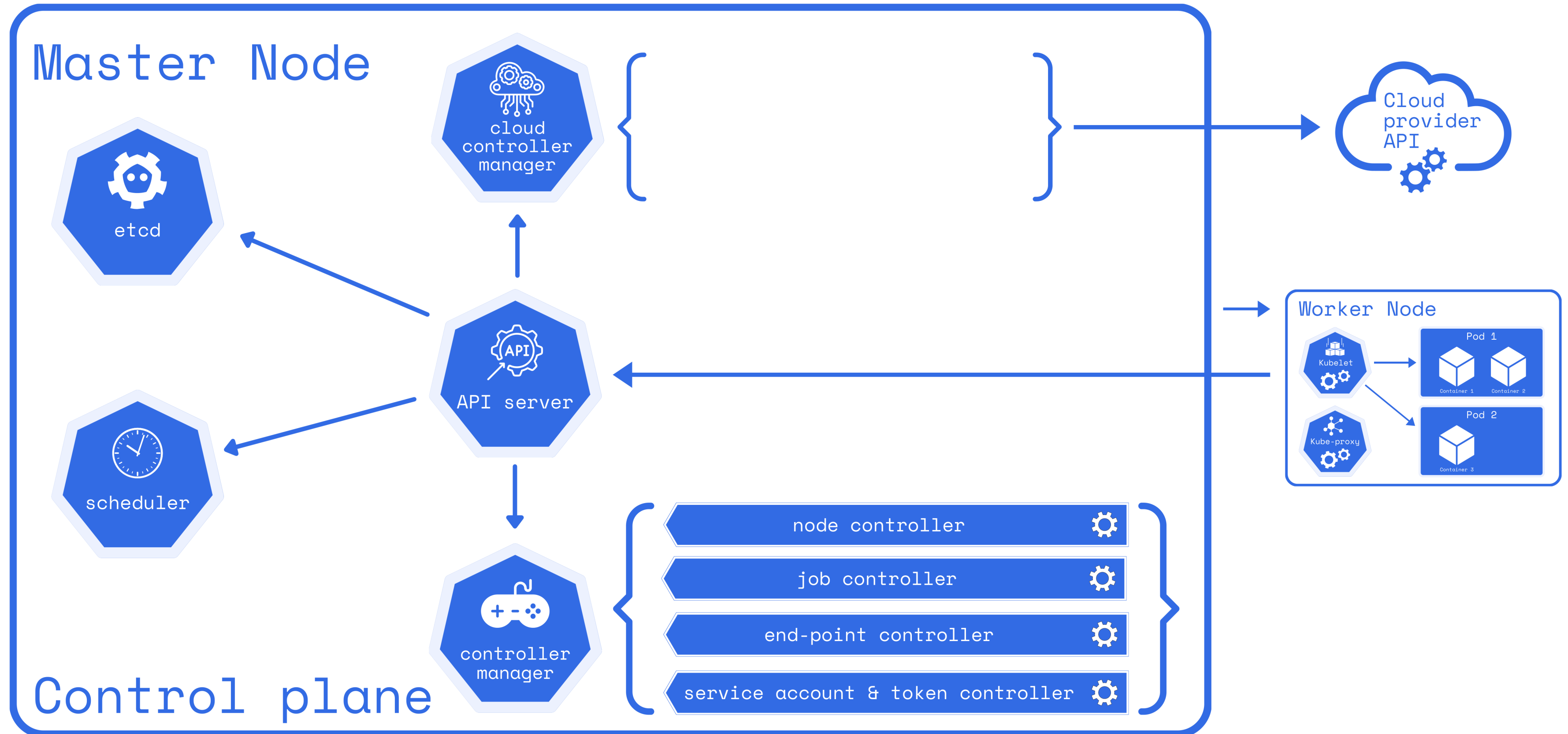
# Kubernetes' components



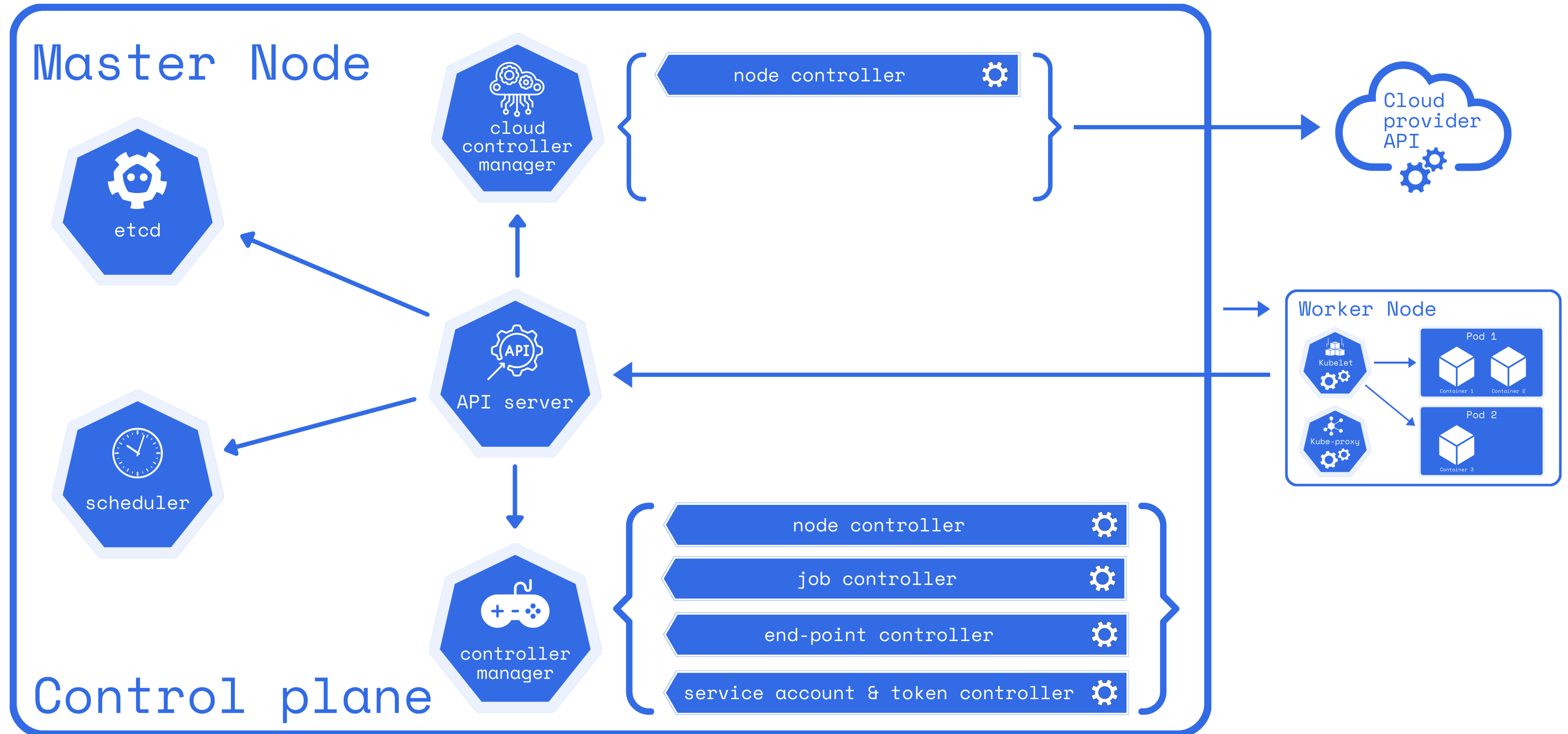
# Kubernetes' components



# Kubernetes' components

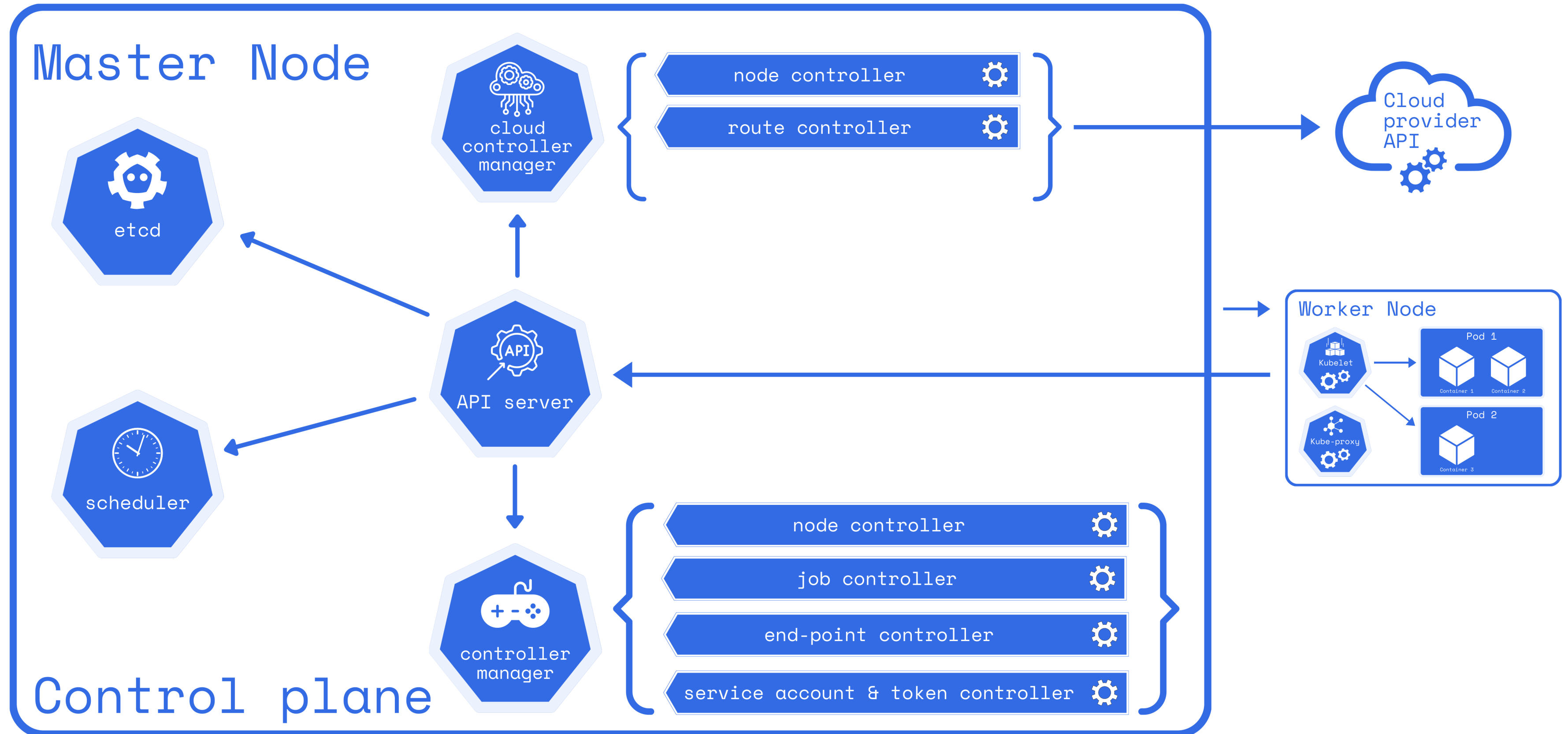


# Kubernetes' components





# Kubernetes' components



# Kubernetes' components

